
Laboratory Guide

Lesson 04 — Discovery and Interrogation of BLE Devices

Communication Technologies for the Internet of Things

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Estimated duration: 2 hours

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Context

The lecture described the link layer, the advertising and scanning states, and the organisation of data into services and characteristics under GATT. This session observes those mechanisms in operation, on real devices, using the BlueZ stack tools presented in the slides.

No software is developed. The commands used are provided and executed in a console, and the student's work consists of interpreting what is returned and relating it to the concepts from the lecture.

Material

- Raspberry Pi with BlueZ installed, acting as the central device.
- One smartphone per group, with the nRF Connect application installed, acting as the BLE peripheral. No additional equipment is required.

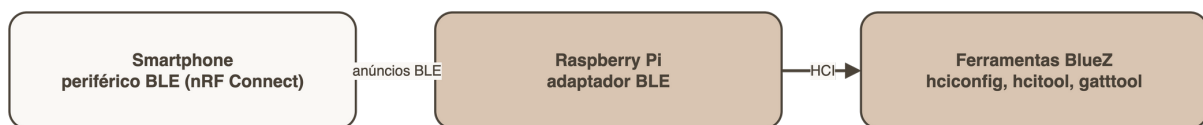


Figure 1: Architecture of the exercise

Learning Outcomes

- Identify BLE devices in the vicinity and distinguish those advertising in a connectable manner from those merely broadcasting information;
- Traverse the GATT hierarchy of a real device, from service down to characteristic value;
- Relate the observed identifiers (UUID and *handle*) to the concepts of attribute type and address presented in the lecture;
- Recognise, in practice, the difference between advertising-based and connection-based communication.

1. Adapter Preparation

15 min

Objective

Confirm the presence of the BLE adapter and bring it to an operational state.

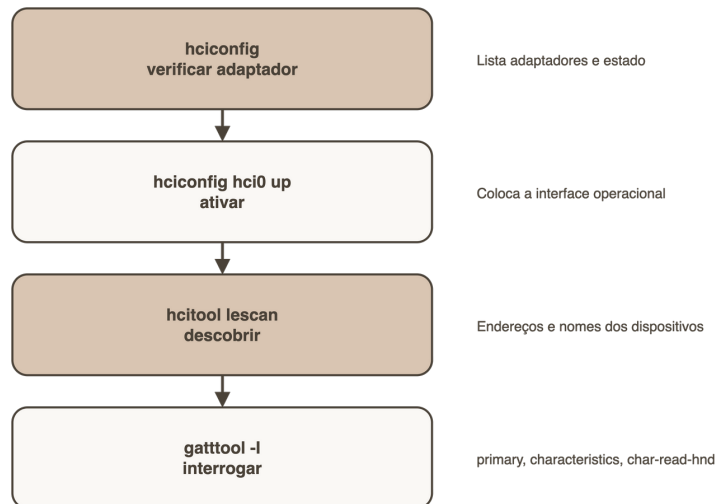


Figure 2: Tool usage sequence

```
hciconfig
sudo hciconfig hci0 up
hciconfig -a hci0
```

1. Run `hciconfig` with no arguments and record the name of the interface shown, usually `hci0`, together with its address.
2. Enable the interface and query the state again with the `-a` option.
3. Record the HCI version reported and compare it with the corresponding Bluetooth specification version.

Question

Is the address shown by `hciconfig` public or random? What consequence does this distinction have for the possibility of tracking the device over time?

2. Device Discovery

30 min

Objective

Observe advertising traffic and characterise the devices present in the room.

```
sudo hcitool lescan
```

1. Run the scan for about 30 seconds and interrupt with **Ctrl+C**.

2. Record in a table the addresses and names obtained. Note how many devices present a name and how many appear with an address only.
3. On the smartphone, open nRF Connect and enable peripheral mode (*Advertiser*), with a name identifiable by the group. Repeat the scan and confirm that the device now appears in the list.
4. Disable advertising on the smartphone and repeat the scan a third time.

Address	Name	Observations
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Relation to the lecture

The scan performed by `hcitool lescan` is active: besides listening on channels 37, 38 and 39, it issues scan requests to which the advertiser replies with additional information. That is why some devices present a name and others do not.

3. Interrogating the GATT Hierarchy

45 min

Objective

Traverse the services and characteristics of a real peripheral and read the value of a characteristic.

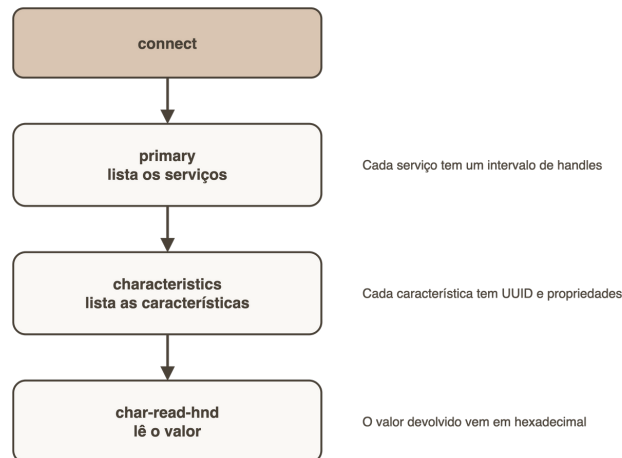


Figure 3: GATT exploration sequence

Keep the smartphone advertising and, from the Raspberry Pi, run:

```
sudo gatttool -I -b <DEVICE_ADDRESS>
```

In interactive mode, issue in sequence:

```
connect
primary
characteristics
char-read-hnd <handle>
disconnect
```

1. Record the services returned by **primary**, noting for each the UUID and the range of *handles*.
2. Record the characteristics returned by **characteristics**, noting the value *handle*, the properties and the UUID.
3. Using **char-read-hnd**, read the value of a characteristic whose properties include reading. The result is shown in hexadecimal.
4. Identify, among the UUIDs observed, those corresponding to services standardised by the Bluetooth SIG and those specific to the manufacturer.

Reading the identifiers

The 16-bit UUIDs, shown in the form 0x180A, correspond to services adopted by the Bluetooth SIG. The 128-bit UUIDs identify services defined by the manufacturer. This distinction makes it immediately apparent which part of the device follows standardised profiles.

Further work

With the nRF Connect application acting as client, connect to a second device and compare the information presented by the application with that returned by `gatttool`. The application automatically resolves standardised UUIDs into readable names; identify which.

4. Advertising and Connection

20 min

Objective

Distinguish experimentally the two communication paradigms studied.

1. Configure nRF Connect to broadcast data in the advertising packet itself, without allowing connection.
2. Confirm by scanning that the device remains visible and that the data is readable with no connection established.
3. Attempt to connect with `gatttool` and record the result.
4. Restore connectable mode and confirm that connection becomes possible.

Questions for Discussion

1. What information was obtainable about third-party devices without any authorisation, and what are the implications of that fact?
2. Why is advertising-based broadcasting more energy-efficient than connection-based communication?
3. Is the activity tracker case presented in the lecture observable in the room? How could it be verified?
4. Which pairing method would be used between the Raspberry Pi and a sensor with neither display nor keyboard, and what risk does that entail?

References

- Gonçalves, P. *Apresent4 — Bluetooth*, TCIoT, previous editions.
- Bluetooth SIG. *Bluetooth Core Specification*.
- `hciconfig` documentation: <https://helpmanual.io/man1/hciconfig/>
- `gatttool` documentation: <https://helpmanual.io/man1/gatttool/>